Electrolytes and the Dissolving Process

Answer key with worked solutions

A. Ideas

1. Definitions.

> (a) **Solute**: the substance dissolved. **Solvent**: the substance it dissolves in. **Electrolyte**: a substance that forms ions in solution, so the solution conducts. (b) Salt separates into **mobile ions** (Na^+ and Cl^-) that carry charge. Sugar dissolves as **whole molecules** with no charge, so nothing carries the current.

2. Classify.

> (a) **strong** (b) **non-electrolyte** (c) **strong** (d) **weak** (e) **strong** (f) **non-electrolyte**

B. Dissociation and ion concentrations

3. Dissociation equations.

> (a) $K_2SO_4\,(s) \longrightarrow 2\,K^+\,(aq) + {SO_4}^{2-}\,(aq)$
> (b) $Ca(NO_3)_2\,(s) \longrightarrow Ca^{2+}\,(aq) + 2\,{NO_3}^-\,(aq)$
> (c) $(NH_4)_3PO_4\,(s) \longrightarrow 3\,{NH_4}^+\,(aq) + {PO_4}^{3-}\,(aq)$
> (d) $FeCl_3\,(s) \longrightarrow Fe^{3+}\,(aq) + 3\,Cl^-\,(aq)$

4. Ion concentrations.

> (a) $[K^+] = 2 \times 0.200\ \text{mol/L} = \mathbf{0.400\ mol/L}$; $[{SO_4}^{2-}] = \mathbf{0.200\ mol/L}$ (b) $[Fe^{3+}] = \mathbf{0.150\ mol/L}$; $[Cl^-] = 3 \times 0.150\ \text{mol/L} = \mathbf{0.450\ mol/L}$

5. Best conductor.

> **$CaCl_2$**: it gives 3 ions per formula unit, 0.30 mol/L of ions in total. NaCl gives 0.20 mol/L of ions; ethanoic acid is only slightly ionized; glucose gives no ions.

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6. Mixing.

> Total volume $= 0.0500\ \text{L} + 0.0500\ \text{L} = 0.1000\ \text{L}$
>
> Ca^{2+}: $0.100\ \text{mol/L} \times 0.0500\ \text{L} = 5.00 \times 10^{-3}\ \text{mol}$, so $[\text{Ca}^{2+}] = \dfrac{5.00 \times 10^{-3}\ \text{mol}}{0.1000\ \text{L}} =$ **$0.0500\ \text{mol/L}$**
>
> K^{+}: $0.200\ \text{mol/L} \times 0.0500\ \text{L} = 1.00 \times 10^{-2}\ \text{mol}$, so $[\text{K}^{+}] =$ **$0.100\ \text{mol/L}$**
>
> Cl^{-}: $2 \times 5.00 \times 10^{-3}\ \text{mol} + 1.00 \times 10^{-2}\ \text{mol} = 2.00 \times 10^{-2}\ \text{mol}$, so $[\text{Cl}^{-}] =$ **$0.200\ \text{mol/L}$**

C. Energy changes

7. Hot or cold.

> Dissolving an ionic solid needs energy to **break up the lattice** (endothermic) and releases energy when the ions are **hydrated** (exothermic). If hydration releases **more** energy than the lattice needs, the overall change is exothermic and the water warms (e.g. CaCl_2). If it releases **less**, the overall change is endothermic and the energy comes from the water, which cools (e.g. NH_4NO_3).

D. Challenge

8. Hand warmer.

> $n = \dfrac{5.00\ \text{g}}{110.98\ \text{g/mol}} = 0.04505\ \text{mol}$
>
> Heat released: $0.04505\ \text{mol} \times 82.8\ \text{kJ/mol} = 3.730\ \text{kJ} = 3730\ \text{J}$
>
> $$\Delta T = \frac{q}{mc} = \frac{3730\ \text{J}}{105\ \text{g} \times 4.184\ \text{J/(g·°C)}}$$
> $$= \mathbf{8.49\ ^\circ\text{C rise}}$$

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